1. **(c)**

Around $x=\frac{2\pi}{3}$, $|\cos x|=-\cos x$ and $|\sin x|=\sin x$

$\therefore\ y=-\cos x+\sin x$ $\therefore\ \frac{dy}{dx}=\sin x+\cos x$

At $x=\frac{2\pi}{3}$, $\frac{dy}{dx}=\sin\frac{2\pi}{3}+\cos\frac{2\pi}{3}$ $=\frac{\sqrt{3}}{2}-\frac{1}{2}=\frac{1}{2}(\sqrt{3}-1)$.

2. **(b)**

Applying Leibnitz's theorem by taking x^2 as second function. We get , $D^n y=D^n(e^x.x^2)$

$={}^nC_0 D^n(e^x)x^2+{}^nC_1 D^{n-1}(e^x).D(x^2)+{}^nC_2 D^{n-2}(e^x).D^2(x^2)+.........$ $\therefore =$ $e^x.x^2+ne^x.2x+\frac{n(n-1)}{2!}e^x.2+0+0+.........$

$y_n=\{x^2+2nx+n(n-1)\}e^x$.

3. **(a)**

Given (2,3) lies on $y^2=px^3+q$ we get 9 = 8p + q……..(1)

Again $y^2=px^3+q$

$\Rightarrow 2y\frac{dy}{dx}=3px^2$

$\Rightarrow \frac{dy}{dx}=\frac{3px^2}{2y}$

$\Rightarrow \left(\frac{dy}{dx}\right)_{(2,3)}=\frac{12p}{6}=2p$

since $y=4x-5$ is tangent to $y^2=px^3+q$ at $(2,3)$ therefore 2p = 4

$[\because \text{Scope of line } y=4x-5 \text{ is } 4]$ $\Rightarrow$ p -2. Putting p=2 in equation (1), we get q= -7

Hence p = 2, q = -7.

4. **(d)**

For y = $\frac{x^2-2}{x^2-4}$, $\frac{dy}{dx}=\frac{-4x}{(x^2-4)^2}$

$\Rightarrow \frac{dy}{dx}>0$ for x < 0 and $\frac{dy}{dx}<0$ for x > 0. Then x = 0 is the point of local maxima for y. Now $y|_{x=0}=\frac{1}{2}$ (positive). Thus x = 0 is also the point of local maxima for $y=\left|\frac{x^2-2}{x^2-4}\right|$

5. **(b)**

When y =2, $e^{-2x}=1 \Rightarrow x=0$,

and $\frac{df}{dx}=-2e^{-2x} \Rightarrow \left.\frac{df}{dx}\right|_{x=0}=-2$

$\Rightarrow$ The equation of the tangent is, (y -2) = -2 (x - 0) $\Rightarrow$ 2 x + y - 2 = 0

6. **(b)**

Differentiating, we get $f'(x)=\left(\frac{\sqrt{a+4}}{1-a}-1\right)5x^4-3$

For $f(x)$ to be decreasing for all x, we must have

$f'(x)<0$ for all x.

$\Rightarrow \left(\dfrac{\sqrt{a+4}}{1-a}-1\right)x^4 < \dfrac{3}{5}\forall x$

This is possible only if $\dfrac{\sqrt{a+4}}{1-a}-1\le 0$

This inequality is always true if $a>1$, i.e., $a\in(1,\infty)$. Moreover, we must have $a\ge -4$ for $\sqrt{a+4}$ to be real. Therefore, we have

$\dfrac{\sqrt{a+4}}{1-a}\le 1 \Rightarrow \sqrt{a+4}\le 1-a$

[$\because$ we consider only a < 1]

$\Rightarrow a+4\le 1+a^2-2a \Rightarrow a^2-3a-3$

$\Rightarrow a\le \dfrac{3-\sqrt{21}}{2}$

Thus, $a\in\left[-4,\dfrac{3-\sqrt{21}}{2}\right]\cup(1,\infty)$

7. (b)

$f(x)=\sin^p x.\cos^q x$

$f'(x)=\sin^p x.q.\cos^{q-1}x.(-\sin x)+\cos^q x.p\sin^{p-1}x.\cos x = \sin^{p-1}x.\cos^{q-1}x\left(p\cos^2 x-q\sin^2 x\right)$.

For maxima & minima $f'(x)=0$

$\Rightarrow \sin^{p-1}x.\cos^{q-1}x\left(p\cos^2 x-q\sin^2 x\right)=0$

$\Rightarrow p\cos^2 x-q\sin^2 x=0$

tanx = $\sqrt{\dfrac{p}{q}}$

$f'(x)=q.\sin^{p-1}x.\cos^{q+1}x.\left(\dfrac{p}{q}-\tan^2 x\right)$

$f'(x-h)=q+ve\left[\dfrac{p}{q}-\left(\tan(x-h)^2\right)\right]=+ve \quad f'(x+h)=+ve\left[\dfrac{p}{q}-\left(\tan(x+h)\right)^2\right]=-ve$

sign changes from +ve to -ve;

so x = $\tan^{-1}\left(\sqrt{\dfrac{p}{q}}\right)$ is point of local maxima.

So, 'a' is correct.

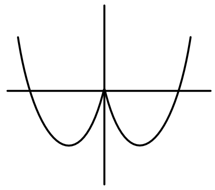
8. (a)

b = 0 , a < 0

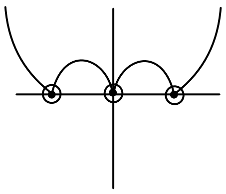
f(x) = x^2 + ax + b is quadratic polynomial cut x axis at x = 0 and x = - a

f(x) = x^2 + ax + b

f (| x |) = x^2 + a |x| + b



$|f|x|| = |x^2 + a|x| + b|$



Exactly at three points function is not differentiable.

9. (c)

$x^y = e^{x-y} \Rightarrow y \log x = x - y$ ....(i)

$\frac{y}{x} + \log x \frac{dy}{dx} = 1 - \frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx}(1+\log x) = 1 - \frac{y}{x}$

$\Rightarrow \frac{dy}{dx} = \left(1-\frac{y}{x}\right) . \frac{1}{(1+\log x)}$

from (i) put value of y

$= \left[1-\frac{x}{(1+\log x)}\times\frac{1}{x}\right]\left(\frac{1}{1+\log x}\right)$

$= \frac{1+\log x-1}{(1+\log x)^2} = (\log x).(1+\log x)^{-2}$

10. (a)

$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{f'(x^2).2x}{9'(x^3).3x^2} = \frac{\sin x^2.2x}{\cos x^3.3x^2}$

$= \frac{2\sin x^2}{3x\cos x^3}$

11. (a)

$f(x) = x^2 + x$ at $x = -2$

$f'(x) = 2x + 1 \Rightarrow f'(-2) = -4 + 1 = -3$

$f(-2) = 4 - 2 = 2$

equation of tangent

$y - 2 = -3(x + 2) \Rightarrow 3x + y + 4 = 0$

12. (d)

Let α, β be two different roots of $f(x) = 0$ in $[0, 1]$ where $f(x) = x^3 - 3x + a$. Therefore, by Rolle's theorem $f'(x) = 0$ has a root between α and β, i.e. in $(0, 1)$. But, $f'(x) = 0 \Rightarrow 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

$\therefore f'(x) = 0$ has no roots in $(0, 1)$

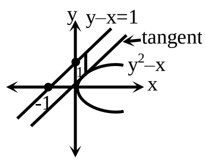
Hence the given equation has no root lying between 0 and 1 for any value of a.

13. (a)

$2y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2y} = 1 \Rightarrow y = \frac{1}{2}$ & $x = \frac{1}{4}$

Now $\perp$ distance from $(1/4, 1/2)$ on

$-x + y - 1 = 0$ is $d = \left|\frac{-1/4+1/2-1}{\sqrt{2}}\right| = 3/4\sqrt{2}$



14. (c)

$f'(x) = 2e^{2x} + (a+1)e^x + 2 \geq 0 \ \forall\ x \in R$

i.e. $2\left(e^x + \frac{1}{e^x}\right) - (a+1) \geq 0 \quad \forall\ x \in R$

$\Rightarrow 4 - (a+1) \geq 0$ or $a \leq 3$

15. (a)

$f'(x) = \frac{ad - bc}{(c\sin x + d\cos x)^2}$ (on solving)

$f'(x)$ is monotonic decreasing if $f'(x) < 0$

$\Rightarrow ad - bc < 0$

16. (b)

$$f'(x) = \frac{\frac{1}{\sqrt{1-x^2}} \times (\cos^{-1}x - \sin^{-1}x) \times \left(-\frac{1}{\sqrt{1-x^2}}\right)}{(\cos^{-1}x)^2}$$

$= \frac{\cos^{-1}x + \sin^{-1}x}{\sqrt{1-x^2}(\cos^{-1}x)^2} = \frac{\pi}{\sqrt{1-x^2}(\cos^{-1}x)^2} = +\text{ve}$

$= f'(x) > 0$ st. Inc. function

17. (a)

$\because f'(x) = m = e^x[\cos x - \sin x]$

$\frac{dm}{dx} = e^x(-\sin x - \cos x) + (\cos x - \sin x)e^x$

$\frac{dm}{dx} = -2\sin x e^x = 0$

$\Rightarrow x = 0, \pi$ ....

$\frac{d^2m}{dx^2} = -2[e^x \cos x + e^x \sin x]$

$\left(\frac{d^2m}{dx^2}\right)_{x=0} = -2 < 0 \Rightarrow$ max. at $x = 0$

18. (d)

$\therefore f(x) = \sin\theta + \cos\theta = \sqrt{2}\ \sin\left(\frac{\pi}{4} + \theta\right)$

maximum value at $\theta = \frac{\pi}{4}$

$\therefore \frac{\{x\}}{a} = \frac{\pi}{4} \Rightarrow \{x\} = \frac{\pi a}{4}$

$\because 0 \leq \frac{\pi a}{4} < 1 \Rightarrow \boxed{0 \leq a < \frac{4}{\pi}}$

$\Rightarrow a \in \left(0, \frac{\pi}{4}\right) \quad \because a \neq 0$

19. (b)

$\frac{dy}{dx} = \frac{a}{x} + 2bx + 1$

since x = - 1, x = 2 are extreme points

so $\frac{dy}{dx} = 0$

$\Rightarrow$ -a - 2b + 1 = 0 & a/2 + 4b + 1 = 0

a + 2b - 1 = 0 & a + 8b + 2 = 0

$\Rightarrow$ a = 2 & b = - 1/2

20. (a)

$\frac{dy}{dx} = \frac{1}{\sqrt{1-\cos^2 x}} \times -\sin x = \frac{-\sin x}{|\sin x|}$

if $x < \frac{\pi}{2}$ |sinx| = sin x

$\Rightarrow \frac{dy}{dx} = \frac{-\sin x}{\sin x} = -1$